

EE 210: Digital System Design

Course Code: EE 210

Course Name: Digital System Design

L-T-P-C: 3-0-0-3

Prerequisites: Applied Electronics (IC 161) and Applied Electronics Lab (IC 161P)

Intended for: B. Tech (EE)

Distribution: Core for B. Tech (EE) & Elective for other UG programs

Semester: Even/Odd

Approval: 24th Senate (recommended by the 32nd BoA; replaces EE 208P Digital Systems Design Practicum together with EE 210P); code corrected from EE 208 to EE 210 in the SCEE minors list by the 30th Senate

Preamble & Objective

A primary objective of this course is to create strong foundation of digital design in our students. This course inculcates deeper understanding of the subject in students that will enable them to design and understand various real-world digital systems. Any subject becomes interesting to student when they gather hierarchical understanding of the topics in detail fashion and then escalate to next-level by applying this knowledge to develop their own system to cater the real-world problems. The content of this course renders exactly the same experience to the students.

Course Content

- **Combinational Logic Design with MSI Components and Programmable Logic Devices:** 1.1 Binary Adders and Subtractors {Cascading Full-Adders, Carry Look-ahead Adder, High-speed Adders using Carry Look-ahead Principles, using MSI Adders as Subtractors, BCD Adder, using MSI adder as code converter}; 1.2 Arithmetic Logic Unit; 1.3 Binary Multipliers; 1.4 Array Multipliers; 1.5 Tristate Buffers; 1.6 Combinational Logic Hazards {Static and Dynamic Hazards}. (4 hours)
- **Flip-Flops and Simple Flip-Flop Applications:** 2.1 Basic Bi-stable Element; 2.2 Application of the SR Latch as Switch De-bouncer; 2.3 Gated SR and D Latch; 2.4 Timing Considerations {Propagation Delays, Contamination Delays, Minimum Pulse Width, Setup and Hold Times}; 2.5 Pulse Triggered Master-Slave Flip-Flops {SR & JK Master-Slave Flip-Flops, 0's and 1's Catching}; 2.6 Edge Triggered Flip-Flops {Positive & Negative Edge Triggered Flip-Flops, Master Slave Flip-flops with Data Lockout}; 2.7 Characteristics Equations; 2.8 Registers; 2.9 Counters; 2.10 Design of Synchronous Counters; 2.11 Self-Correcting Counters. (5 hours)
- **Synchronous Sequential Networks and Algorithmic State Machine (ASM):** 3.1 Introductions to State Equivalence; 3.2 State Reductions {Equivalence Classes and Implication Charts}; 3.3 State Reduction of Incompletely Specified State Table using Merger Graphs; 3.4 State Assignment Techniques {State Assignment Permutations, State Assignment Algorithm, Implication Graph}; 3.5 Algorithm State Machine {ASM Symbols, Elapse Time Measurement as an ASM Design Example}; 3.6 Linked Sequential Machines. (3 hours)
- **Asynchronous Sequential Networks:** 4.1 Fundamental and Pulse Mode Asynchronous Sequential Machines; 4.2 Analysis of Asynchronous Sequential Machines; 4.3 Deriving Flow Table; 4.4 State Assignment; 4.5 Asynchronous Design Problems; 4.6 Data Synchronizers; 4.7 Mixed Operating Mode Asynchronous Circuits. (7 hours)

- **Programmable Logic and Memory:** 5.1 Introductions to Memory {ROM, PROM and EPROM}; 5.2 Using and EPROM to Realize a Sequential Circuit; 5.3 Programmable Logic Devices {PLA, PAL, GAL}; 5.4 Erasable Programmable Logic Devices; 5.5 PLD Computer-Aided Design {PLD Realization of Combinational Logic, Realization of truth table, flip-flops and state machine using PLD language}. (5 hours)
- **Hardware Description Language and Field-Programmable Gate-Array (FPGA):** 6.1 Introductions to Hardware Description Language (HDL); 6.2 Overview of Structural, Behavioral and Dataflow Modeling of Digital Systems using HDL; 6.3 HDL Realizations of Finite State Machines, Delay Modeling, Memory Modeling; 6.4 Synthesizable & Non-Synthesizable HDL Codes for Digital System Design; 6.5 Overview of FPGA architecture; 6.6 Realization of Data-Path and Controller; 6.7 Timing Analysis of Data-Path and Controller; 6.8 Synthesis, Placement, Routing and Performance Optimization; 6.9 Implementation of Simple Digital Systems using FPGA Exercising the Timing Closure Paths. (7 hours)
- **Microcontroller and Real-World Applications:** 7.1 Introduction to microcontrollers; 7.2 Architectural Overview of a Typical Microcontroller such as AVR Microcontroller; 7.3 Addressing; 7.4 Assembly Language Programming; 7.5 Interfacing with I/O devices, timer/counter programming, interrupt processing etc; 7.6 Interfacing with Simple Devices such as LCD, Keyboard, Motor Control, Sensors, LED 7 Segment Display, DTMF decoder etc; 7.7 SPI, I²C, Programmable Interrupt Controller, USART etc. (9 hours)

Text Books

1. Donald D. Givone, *Digital Principles and Design*, Tata McGraw-Hill Edition (2012).
2. John M. Yarbrough, *Digital Logic (Applications and Design)*, Cengage Learning (2011).

Reference Books

1. Dhananjay Gadre, *Programming and Customizing the AVR microcontroller*, Tata McGraw Hill, 2014.
2. Wayne Wolf, *FPGA based Systems Design*, Pearson Education, 2003.
3. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital with VHDL Design*, McGraw Hill, 2008.
4. Volnei A. Pedroni, *Circuit Design with VHDL*, The MIT Press, 2004.
5. Steve Kilts, *Advanced FPGA Design: Architecture, Implementation and Optimization*, J. Wiley and Sons, 2007.
6. Seetharaman Ramachandran, *Digital VLSI Systems Design*, Springer Verlag, 2012.
7. Peter J. Ashenden, *The designer's guide to VHDL*, Morgan Kaufmann, 2008.
8. Charles H. Roth Jr., *Digital Systems Design using VHDL*, Cengage Learning, 2014.